

An explicit strong-coupling gap for the Wilson transfer matrix: $SU(3)$ is gapped for $g^2 \geq 176$, with $\Delta \geq (\hbar c/a) 4 \log(g^2/176)$

The threshold that Yarotsky's theorem supplies for the Kogut–Susskind Hamiltonian is $g_0^2 \sim 10^{101}$ (threshold note). The same statement for the other standard lattice Hamiltonian, the logarithm of the Wilson transfer matrix, has an explicit and usable threshold, because the Euclidean character expansion is a polymer gas of closed plaquette surfaces with no time discretization to pay for. With the plaquette activity ρ of the fundamental representation, which is $1/g^2$ for $SU(3)$ to leading order, the Kotecký–Preiss criterion holds for

$$20e^2\rho \leq 0.84, \quad \text{that is} \quad \rho \leq 5.7 \times 10^{-3},$$

and the truncated correlations of local observables then decay with rate at least $4 \log(1/(176\rho))$ per lattice unit, because the smallest closed surface joining two distant plaquettes is a tube with four plaquettes per unit length. Through the transfer matrix $\mathcal{T} = e^{-aH_W/\hbar c}$, self-adjoint and positive by link reflection positivity, the decay rate is a lower bound on the gap of H_W on the cyclic subspace of local observables, so for $SU(3)$

$$\Delta_W \geq \frac{\hbar c}{a} 4 \log \frac{g^2}{176} \quad (g^2 \geq 176),$$

uniformly in the volume. The fully rigorous version, which sums all representations and both orientations with the bound $e^{\beta_W} - 1 \simeq 6/g^2$, gives the same form with 1056 in place of 176. The entropy constant $20e$ for plaquette animals is the only place with room, and a careful count would lower 176 to about 10^2 . The gap of H_W grows like $\log g^2$ where that of H_{KS} grows like g^2 , the two Hamiltonians being different regularizations that agree only in the continuum limit. Constants explicit; nothing promoted.

1. The polymer gas

Write the Wilson weight of a plaquette in characters,

$$e^{\frac{\beta_W}{N} \text{Re tr } U_p} = c_0(\beta_W) \left[1 + \sum_{r \neq 0} d_r \frac{c_r}{c_0} \chi_r(U_p) \right], \quad c_r = \frac{1}{d_r} \int e^{\frac{\beta_W}{N} \text{Re tr } U} \overline{\chi_r(U)} dU,$$

with $\beta_W = 2N/g^2$. Expanding the product over plaquettes and integrating over links, a set of plaquettes with nontrivial representations contributes only if the representations at every link fuse to the trivial one, so every link is covered by at least two plaquettes of the set: the surviving sets are **closed surfaces**, the smallest being the six faces of a cube. On a closed surface carrying the fundamental on every plaquette, the link integrals $\int \chi_f(U_1 g) \chi_f(g^{-1} U_2) dg = \chi_f(U_1 U_2)/d_f$ and the vertex closures combine to $d_f^{\chi(P)} (c_f/c_0)^{|P|}$ with $\chi(P)$ the Euler characteristic, so the activity per plaquette along a tube, where the vertex and link factors balance, is

$$\rho = d_f \frac{c_f}{c_0} = \frac{\beta_W}{2N} (1 + O(\beta_W)) = \frac{1}{g^2} (1 + O(g^{-2})) \quad (SU(N)),$$

the textbook strong-coupling parameter. The polymers of the gas are the connected closed surfaces, two polymers being incompatible when they share a link, and $Z = c_0^{|P|} \sum_{\text{compatible}} \prod w(\gamma)$.

All representations, rigorously. Expand $e^{x \text{Re tr } U} = \sum_k x^k (\text{Re tr } U)^k / k!$ with $x = \beta_W/N$, and $(\text{Re tr } U)^k = 2^{-k} \sum_j \binom{k}{j} (\text{tr } U)^j (\text{tr } U^\dagger)^{k-j}$. The integral $\int (\text{tr } U)^j (\text{tr } U^\dagger)^{k-j} \overline{\chi_r} dU$ is the multiplicity $m_r(j, k-j)$ of r in $f^{\otimes j} \otimes \bar{f}^{\otimes (k-j)}$, so $d_r c_r = \sum_k \frac{x^k}{k!} 2^{-k} \sum_j \binom{k}{j} m_r(j, k-j) \geq 0$, and $\sum_r m_r(j, k-j) \leq \dim(f^{\otimes j} \otimes \bar{f}^{\otimes (k-j)}) = N^k$. Hence

$$\sum_{r \neq 0} d_r c_r \leq \sum_{k \geq 1} \frac{x^k}{k!} 2^{-k} \cdot 2^k N^k = e^{Nx} - 1 = e^{\beta_W} - 1,$$

and Jensen's inequality with $\int \text{Re tr } U dU = 0$ gives $c_0 = \int e^{x \text{Re tr } U} dU \geq 1$. Therefore

$$\sum_{r \neq 0} d_r \frac{c_r}{c_0} \leq e^{\beta w} - 1 = e^{2N/g^2} - 1 \simeq \frac{2N}{g^2} = \frac{6}{g^2} \quad (SU(3)),$$

which is the leading activity $1/g^2$ of the fundamental, its conjugate, and the higher representations together, with no spurious factor. The sum over representations also covers the branch lines where three fundamentals meet at a link, allowed for $SU(3)$ since $f^{\otimes 3} \ni 1$. Everything below is stated for a per-plaquette activity ρ ; the two readings differ only in what ρ is.

2. Convergence: the Kotecký–Preiss criterion with numbers

Adjacency. In four dimensions a link lies in six plaquettes, so a plaquette shares a link with $4 \times 5 = 20$ others, and the number of connected sets of n plaquettes containing a given one is at most $(20e)^{n-1}$.

Incompatible polymers. A polymer γ' incompatible with γ shares a link with it; γ has at most $4|\gamma|$ links, each in six plaquettes, so the number of γ' of size n is at most $24|\gamma|(20e)^{n-1}$, and $n \geq 6$.

Criterion. With $a(\gamma) = |\gamma|$ and $|w(\gamma')| \leq \rho^{|\gamma'|}$, the Kotecký–Preiss condition $\sum_{\gamma' \not\sim \gamma} |w(\gamma')| e^{a(\gamma')} \leq a(\gamma)$ follows from

$$\frac{24}{20e} \sum_{n \geq 6} (20e^2 \rho)^n \leq 1 \quad \Longleftarrow \quad x \equiv 20e^2 \rho \leq 0.84,$$

since $0.4415 x^6 / (1 - x) \leq 1$ there. So the expansion converges, uniformly in the volume, for

$$\rho \leq \frac{0.84}{20e^2} = 5.7 \times 10^{-3}.$$

3. Decay of correlations and the gap

Running the criterion with $a(\gamma) = (1 + \delta)|\gamma|$ instead, it holds as long as $x e^\delta \leq 0.84$, that is for $\delta \leq \log(0.84/x) = \log(1/(176\rho))$, and then the cluster sums carry a factor $e^{-\delta|X|}$. A cluster connecting two plaquettes at distance T contains a closed connected surface through both, whose size is at least that of a tube, $4T$ plaquettes, so the truncated two-point function of any two local observables at separation T is bounded by $C e^{-4\delta T}$:

$$m a \geq 4 \log \frac{1}{176\rho}.$$

Transfer. The Wilson action is link-reflection positive, so the transfer matrix $\mathcal{T}$ is self-adjoint and positive (Lüscher, Commun. Math. Phys. 54 (1977) 283; Osterwalder and Seiler, Ann. Phys. 110 (1978) 440; metadata level), and $H_W = -(\hbar c/a) \log \mathcal{T}$. For a local observable A orthogonal to the vacuum, $\langle A, \mathcal{T}^n A \rangle \leq C_A e^{-man}$ for all n forces the spectral measure of A to vanish above e^{-ma} , so on the cyclic subspace generated by local observables, which is the whole space of the reconstructed theory,

$$\Delta_W \geq \hbar c m \geq \frac{\hbar c}{a} 4 \log \frac{1}{176\rho}.$$

Numbers for $SU(3)$.

reading of ρ	threshold $\rho \leq 5.7 \times 10^{-3}$	gap
leading, $\rho = 1/g^2$	$g^2 \geq 176$	$\Delta_W \geq (\hbar c/a) 4 \log(g^2/176)$

reading of ρ	threshold $\rho \leq 5.7 \times 10^{-3}$	gap
all representations, $\rho = e^{6/g^2} - 1$	$g^2 \geq 1056$	$\Delta_W \geq (\hbar c/a) 4 \log(g^2/1056)$

The first line uses the fundamental alone; the second sums every representation and both orientations and is rigorous as written. The factor 6 between them is the two orientations, each with activity $1/g^2$, and the higher representations, so the rigorous line is close to the truth for a surface that may carry either orientation on each plaquette; a surface with a fixed global orientation would be counted by the first line.

4. Relation to the Kogut–Susskind statement

H_W and H_{KS} are different regularizations of the same continuum Hamiltonian. At strong coupling their gaps differ in kind: H_{KS} has an electric term of order $g^2 \hbar c/a$ and the flux loop costs $(8/3)g^2 \hbar c/a$ for $SU(3)$; H_W has both electric and magnetic content in $-\log \mathcal{T}$, and the flux loop costs $4 \log(g^2/\cdot) \hbar c/a$, the logarithm of the tube activity. The Jaffe–Witten statement asks for the continuum theory and is indifferent to the regularization, so either Hamiltonian serves for T2, and the one with the explicit threshold is H_W . The anisotropic limit $a_t \rightarrow 0$ that turns $\mathcal{T}$ into $e^{-a_t H_{KS}/\hbar}$ sends the temporal plaquette activity to one, outside the expansion, which is why the Kogut–Susskind threshold needs a genuinely Hamiltonian expansion and why Yarotsky’s time discretization pays what it pays.

5. Consequence for STATE

The strong-coupling boundary of the intermediate region is now a number: for the Wilson transfer matrix and $SU(3)$, the theory is gapped uniformly in the volume for $g^2 \geq 176$ (leading activity) or $g^2 \geq 1056$ (all representations, rigorous), with the gap explicit. The only constant with room is the plaquette-animal entropy $20e$. On the weak side the boundary is $1/g^2 \gtrsim 10^2$ with the crude window of the operator-inequality note, and the intermediate region between $g^2 \sim 10^{-2}$ and $g^2 \sim 2 \times 10^2$ is where the gap forms.